
REINFORCE++: Stabilizing Critic-Free Policy Optimization with Global Normalization

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Abstract

Reinforcement Learning from Human Feedback (RLHF) plays a crucial role in aligning Large Language Models (LLMs). The dominant algorithm, Proximal Policy Optimization (PPO), employs a critic network to estimate advantages, which introduces significant computational and memory overhead. To address this, a family of critic-free algorithms (e.g., GRPO, RLOO) has emerged. However, these methods typically rely on *prompt-level (local)* advantage normalization, which suffers from inaccurate advantage estimation, a tendency to overfit, and, as we show, is a theoretically biased estimator. To solve these challenges, we introduce **REINFORCE++**, a critic-free framework centered on **Global Advantage Normalization**. By normalizing advantages across the entire global batch rather than small, prompt-specific groups, our method yields a more stable estimator and, under standard assumptions, enjoys asymptotically vanishing normalization-induced bias as batch size increases. We introduce two variants: **REINFORCE++**, a computationally efficient algorithm ($k \geq 1$) for general-domain RLHF, and **REINFORCE++_{w/ Baseline}**, a group-sampling variant ($k > 1$) for complex reasoning tasks. Across our evaluated settings, each variant improves training stability and matches or exceeds strong critic-free baselines; in complex agentic settings, REINFORCE++_{w/ Baseline} is competitive with PPO.

1. Introduction

Reinforcement Learning from Human Feedback (RLHF) is a key technique for aligning Large Language Models (LLMs) with human values and preferences (Vemprala et al., 2023;

Achiam et al., 2023; Ouyang et al., 2022; Shen & Zhang, 2024; Shen et al., 2025; Hu et al., 2024). Despite the emergence of non-RL alternatives like DPO (Rafailov et al., 2023), widely deployed systems such as ChatGPT/GPT-4 (Vemprala et al., 2023; OpenAI, 2023), Claude (Anthropic, 2023), and Gemini (Team et al., 2023) continue to rely on RL algorithms.

The dominant RL algorithm in this space is Proximal Policy Optimization (PPO) (Schulman et al., 2017). PPO employs an "Actor-Critic" architecture, where a dedicated critic network is trained to estimate the advantage function (i.e., the extent to which an action yields a higher expected return than the mean). However, this critic network introduces substantial computational overhead and memory demands, making training very expensive and limiting large model alignment in small-scale clusters (Shao et al., 2024).

To address PPO’s efficiency issues, a family of REINFORCE-based methods has emerged without the critic model, including ReMax (Li et al., 2023), RLOO (Ahmadian et al., 2024), and GRPO (Shao et al., 2024). These algorithms remove the critic network and instead estimate the advantage using statistics from multiple responses to the same prompt.

However, this critic-free approach introduces a new, critical challenge: *advantage estimation*. Methods like GRPO and RLOO use **prompt-level (local) normalization**, calculating the baseline and standard deviation only from the small group of responses generated for a single prompt. This approach suffers from several critical flaws. First, as demonstrated in Appendix Appendix A, the prompt-level advantage estimator is mathematically biased, as the numerator (centered reward) and the denominator (local standard deviation) are not independent. Second, the advantage estimate is sensitive to the small number of samples in the local group (e.g., $k = 4$ or $k = 8$), where the high variance leads to instability. If all samples receive similar rewards, the local $\text{std}(\cdot)$ approaches zero, causing the advantage to explode and destabilizing training. Finally, the method encourages overfitting to specific prompts, as the policy is optimized to "win" within its local group rather than achieving a globally high reward, which harms generalization.

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We introduce two distinct variants of this framework, each tailored to a specific use case. REINFORCE++ is an efficient algorithm with $k \geq 1$. In its $k = 1$ configuration (detailed in Algorithm 1), it maximizes prompt diversity for general-domain RLHF. It can also be applied with $k > 1$ (Section 4.2), using the same global normalization principle. REINFORCE++_{w/ Baseline} is a specialized *variant* for complex tasks ($k > 1$) that benefits from **group sampling**. It fixes the flaws of GRPO by first subtracting the *group mean* (for reward reshaping) and then normalizing by the *global standard deviation* (for stability). It also employs a more stable k_2 for the KL estimator, as detailed in Appendix Appendix B. In summary, our contributions are as follows:

- We provide a theoretical proof that the prompt-level (local) advantage normalization used in methods like GRPO is a biased estimator (Appendix Appendix A).
- We propose REINFORCE++, a critic-free framework centered on Global Advantage Normalization, as a stable, efficient, and theoretically sound alternative.
- We present two specialized algorithms: We can use REINFORCE++ when $k \geq 1$ for efficient general RLHF and reasoning, and use REINFORCE++_{w/ Baseline} when $k > 1$ for robust complex agentic tasks.
- Our experiments show that REINFORCE++ improves token-efficiency and generalization on out-of-distribution tasks. Meanwhile, REINFORCE++_{w/ Baseline} mitigates overfitting in low-data regimes and performs strongly relative to GRPO and PPO on complex agentic reasoning benchmarks.

2. Background and Related Work

2.1. PPO and Critic-Free RLHF

PPO optimizes LLMs by maximizing the following surrogate objective:

$$\mathcal{L}_{\text{PPO}}(\theta) = \mathbb{E}_{q \sim P(Q), o \sim \pi_{\theta_{\text{old}}}(O|q)} \left[\frac{1}{|o|} \sum_{t=1}^{|o|} \min \left(s_t(\theta) A_t, \text{clip}(s_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t \right) \right] \quad (1)$$

where $s_t(\theta) = \frac{\pi_{\theta}(o_t|q, o_{<t})}{\pi_{\theta_{\text{old}}}(o_t|q, o_{<t})}$ is the probability ratio. The advantage A_t is typically calculated using Generalized Advantage Estimation (GAE) (Schulman et al., 2018):

$$A_{q, o_t} = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l} \quad (2)$$

where $\delta_{q, o_t} = r_t + \gamma V(o_{t+1}) - V(o_t)$ is the temporal difference error, and $V(\cdot)$ is the critic network.

Critic-free methods in Figure 1 remove the critic $V(\cdot)$ and compute the advantage A_t directly from rewards.

- **ReMax** (Li et al., 2023) uses a greedy decoding response $\hat{o}$ as the baseline: $A_{q, o_t} = r(o) - r(\hat{o})$.
- **RLOO** (Ahmadian et al., 2024) samples k responses and uses the mean of others as the baseline: $A_{q, o_t^{(i)}} = r(o^{(i)}) - \frac{1}{k-1} \sum_{j \neq i} r(o^{(j)})$.
- **GRPO** (Shao et al., 2024) also samples k responses but normalizes the advantage using the mean and standard deviation of the *local group*:

$$A_{q, o_t^{(i)}} = \frac{r(o^{(i)}) - \text{mean}(\{r(o^{(j)})\}_{j=1}^k)}{\text{std}(\{r(o^{(j)})\}_{j=1}^k) + \epsilon} \quad (3)$$

2.2. The Problem with Local Normalization

The core issue with methods like GRPO lies in Eq. (3). The usage of **prompt-level (local) normalization** is problematic for three reasons:

1. **Theoretical Bias:** As shown in Appendix Appendix A (under mild distributional assumptions), the estimator is biased for finite group sizes. Intuitively, the numerator (centered reward) and the denominator (local standard deviation $\text{std}(\cdot)$) are statistically coupled within the same small group, which introduces a systematic distortion in the resulting advantage estimate (Mai et al., 2025).
2. **Practical Instability:** In practice, the group size k is small (e.g., 4 or 8). If all sampled responses for a prompt happen to get similar rewards, the local $\text{std}(\cdot)$ approaches zero, causing the advantage to explode. It results in high variance and unstable training.
3. **Task Overfitting:** The policy is rewarded for being "better than other samples from the same prompt," not for being "globally good." It can lead to overfitting on simple prompts where it's easy to generate diverse rewards, while failing to improve on complex prompts.

3. Method

Our method addresses the instability of critic-free RLHF by replacing biased local normalization with stable global normalization. We propose two variants for different use cases.

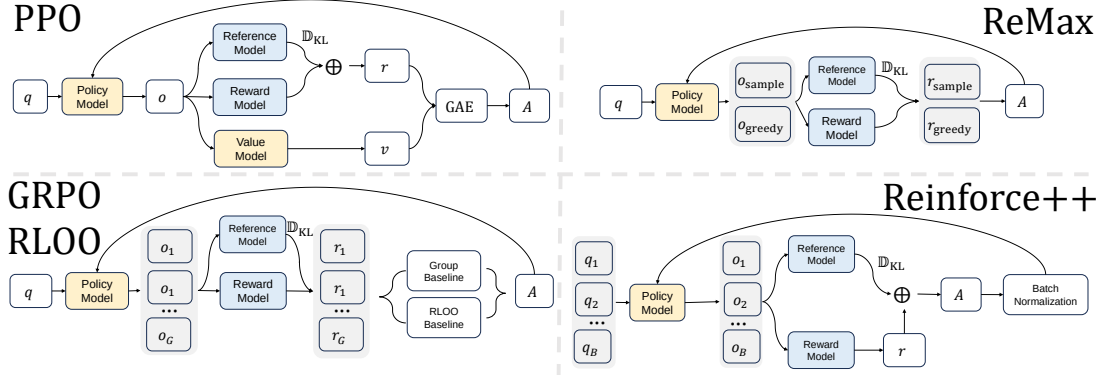


Figure 1. The comparison of PPO, ReMax, GRPO, RLOO, and REINFORCE++. REINFORCE++, as shown in the figure, removes the critic model and uses global batch normalization. REINFORCE++_{w/ Baseline} uses group sampling, similar to GRPO/RLOO, but replaces the local group baseline with a group-mean subtraction followed by global batch normalization.

3.1. Global Normalization

The primary REINFORCE++ algorithm is designed for general-purpose RLHF. In its $k = 1$ configuration, it is designed for maximum efficiency and prompt diversity.

It optimizes the PPO objective in Eq. (1) but redefines the advantage A_{q,o_t} . We use the standard PPO reward formulation, which incorporates the kl-style KL penalty directly into the reward:

$$A_{q,o_t} = r(o_{1:T}, q) - \beta \cdot \sum_{i=t}^T \text{KL}(i) \quad (4)$$

where $\text{KL}(t) = \log \left(\frac{\pi_{\theta_{\text{old}}}(o_t | q, o_{<t})}{\pi_{\text{ref}}(o_t | q, o_{<t})} \right)$.

The key innovation is our normalization strategy. Instead of a non-existent (for $k = 1$) or biased local norm, we use Global Advantage Normalization (Andrychowicz et al., 2020):

$$A_{q,o_t}^{\text{norm}} = \frac{A_{q,o_t} - \text{mean}(A | A \in \mathcal{D}_{\text{batch}})}{\text{std}(A | A \in \mathcal{D}_{\text{batch}}) + \epsilon} \quad (5)$$

This global normalization is the core of REINFORCE++. As the global batch size $\mathcal{D}_{\text{batch}}$ is typically large (e.g., 1024 or more), the empirical mean($\cdot$) and std($\cdot$) are more stable than prompt-level statistics. Under standard assumptions, the normalization-induced coupling diminishes as N grows, leading to an asymptotically vanishing bias and improved robustness to outliers (see Appendix Section 3.5). The algorithm for the efficient $k = 1$ case is detailed in Algorithm 1.

3.2. Local Baseline

For more complex tasks, such as multi-step reasoning, sampling multiple responses per prompt ($k > 1$) can be beneficial (Yue et al., 2025). For this, we introduce

REINFORCE++_{w/ Baseline}. This variant combines the benefits of group-sampling with the stability of global normalization. The advantage calculation is a two-step process:

- **Group Mean Subtraction for Reshaping:** We first subtract the group mean reward. It serves as a local baseline, reshaping rewards to be robust to different reward scales (e.g., 0/1 vs. $-1/1$).

$$A'_{q,o_t} = R_{q,o_t} - \text{mean}_{\text{group}}(R_{q,o_t}) \quad (6)$$

- **Global Batch Normalization for Stability:** We then normalize this initial advantage using the **global batch statistics**, not the unstable local group statistics.

$$A_{q,o_t}^{\text{norm}} = \frac{A'_{q,o_t} - \text{mean}_{\text{batch}}(A')}{\text{std}_{\text{batch}}(A') + \epsilon} \quad (7)$$

This combination addresses the bias and instability introduced by local standard deviation in GRPO.

Furthermore, this variant employs a separate KL loss term for regularization. We adopt the k_2 estimator. As shown in Appendix Appendix B, the k_2 loss yields a stable gradient estimator that matches the reverse-KL gradient in expectation under the sampling distribution, unlike the k_3 estimator (used in GRPO), which can be unstable as an approximation when sampling from π_{θ} (Liu et al., 2025a). The final objective is:

$$\mathcal{L} = \mathcal{L}_{\text{PPO}}(A^{\text{norm}}) - \lambda \cdot \mathcal{J}_{k_2 \text{ as loss}}(\theta) \quad (8)$$

where $\mathcal{J}_{k_2 \text{ as loss}}(\theta) = \mathbb{E}[\frac{1}{2}(\log \frac{\pi_{\theta}}{\pi_{\text{ref}}})^2]$.

3.3. Relationship with PPO

REINFORCE++_{w/ Baseline} can be viewed as a simplified and more stable variant of PPO. It is formally equivalent to a PPO agent where: (1) The critic network is removed; (2)

The GAE parameters are set to $\lambda = 1$ and $\gamma = 1$; and (3) A two-step global batch normalization is used as the baseline instead of a learned value function.

3.4. Implementation Tricks

Token-level Advantage The advantage A_{q,o_t}^{norm} is computed at the token level. For $t < T$ (where T is the sequence length), the advantage is set to 0. For the final token $t = T$, the advantage is the normalized reward A_{q,o_T}^{norm} . This is standard practice in RLHF for LLMs.

Batch Construction For REINFORCE++ ($k = 1$), a global batch of size N consists of N different prompts. For REINFORCE++_{w/ Baseline} ($k = 4$), a global batch of size $N = 1024$ might contain $1024/4 = 256$ unique prompts. The global mean/std are computed over all N samples.

Mini-Batch Updates To enhance training efficiency, we implement mini-batch updates with the following characteristics:

- **Batch Processing:** Data is processed in smaller, manageable chunks rather than full-batch updates.
- **Multiple Updates:** Each mini-batch allows for multiple parameter updates, improving convergence rates.
- **Stochastic Optimization:** Introduces beneficial randomness for better generalization.

Reward Normalization and Clipping We implement comprehensive reward processing to stabilize training:

- **Normalization:** Standardizes rewards using z-score normalization (our global normalization) to mitigate outliers.
- **Clipping:** Constrains reward values within predefined bounds to avoid instability.
- **Scaling:** Applies appropriate scaling factors for numerical stability during updates.

3.5. Why Use Global Batch Normalization?

We observe that as $N \rightarrow \infty$, the batch standard deviation concentrates around the population standard deviation, and the coupling between the centered numerator and the denominator diminishes. This insight underlies our decision to adopt global batch normalization in both REINFORCE++ and REINFORCE++_{w/ BASELINE}, as the global batch size (N_{global}) is typically much larger (e.g., 1024) than the group batch size (N_{group} , e.g., 4 or 8) in practice. As $N_{global} \rightarrow \infty$, the sample statistics μ_b and σ_b converge to constants, and the resulting estimator approaches an asymptotically unbiased form.

3.6. Summary

To address these challenges, we propose REINFORCE++, a method centered on a simple yet powerful idea, **Global Advantage Normalization**, without a critic model. Instead of normalizing within a prompt’s local group, REINFORCE++ normalizes the advantage function across the *entire global training batch*. This approach is:

- **Theoretical Stability:** As the global batch size grows (e.g., $N = 1024$), the batch mean and standard deviation converge to constants; under standard assumptions, the normalization-induced coupling diminishes and the estimator approaches an asymptotically unbiased form (see Appendix Section 3.5).
- **Training Efficiency:** It retains the critic-free architecture, significantly reducing computational and memory overhead compared to PPO.
- **Strong Generalization:** Using a global baseline prevents overfitting to specific prompts and encourages a more robust policy.

Algorithm 1 REINFORCE++ when $k = 1$

Require: Initial policy model π_{ref} , reward models R , task prompts $\mathcal{D}$

- 1: policy model $\pi_\theta \leftarrow \pi_{ref}$
- 2: **for** step = 1, ..., M **do**
- 3: Sample a batch $\mathcal{D}_{batch}$ from $\mathcal{D}$
- 4: Update the old policy model $\pi_{old} \leftarrow \pi_\theta$
- 5: **Sample output** ($k = 1$) $o \sim \pi_{old}(\cdot | q)$ for each question $q \in \mathcal{D}_{batch}$
- 6: Compute rewards r_i for each sampled o_i using R
- 7: Compute advantage A_{q,o_t} via Eq. (4)
- 8: Normalize advantages globally across $\mathcal{D}_{batch}$ to get A_{q,o_t}^{norm} via Eq. (5)
- 9: **for** iteration = 1, ..., k **do**
- 10: Update π_θ by maximizing the objective in Eq. (1) using A_{q,o_t}^{norm}
- 11: **end for**
- 12: **end for**

Ensure: π_θ

4. Experiments

We evaluate our two algorithms, REINFORCE++ and REINFORCE++_{w/ Baseline}, in their respective target domains using the OpenRLHF framework (Hu et al., 2024).

4.1. General RLHF

First, we evaluate the REINFORCE++ in its $k = 1$ configuration, focusing on general-domain RLHF, where efficiency and prompt diversity are key.

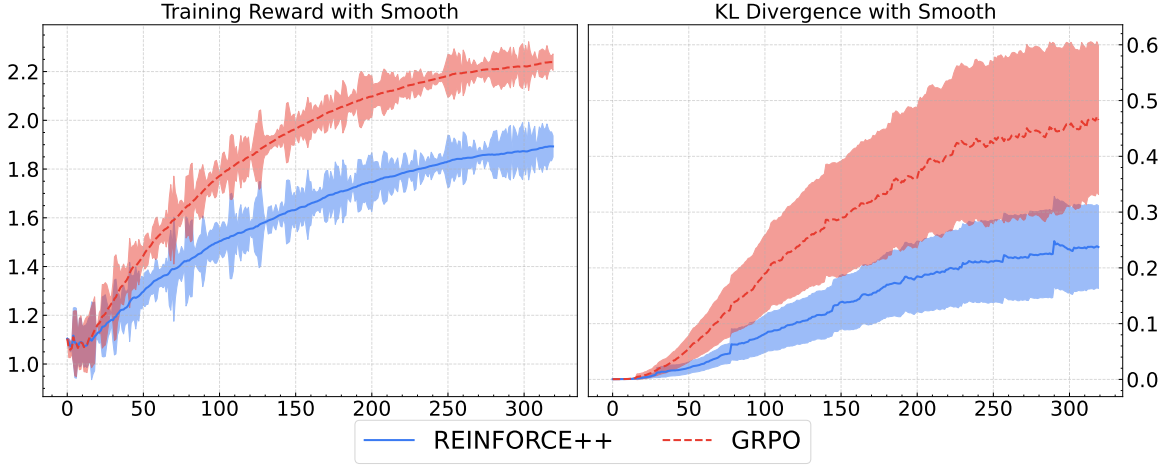


Figure 2. Comparison of smoothed Training Reward and KL Divergence. REINFORCE++ ($k = 1$) achieves strong rewards with more stable (lower) KL divergence, while GRPO exhibits a faster KL increase consistent with reward over-optimization.

Experimental Setup We use an instruction-following policy model (Llama-3-8B-SFT) and refine it using a Bradley-Terry reward model (Bai et al., 2022) trained on $\sim 700K$ human preference pairs. The policy is trained on 20,000 diverse prompts. We compare REINFORCE++ ($k = 1$) with other critic-free methods: GRPO ($k = 4$), RLOO ($k = 4$), and ReMax ($k = 1 + 1$).

	Score	Length	Per Token
REINFORCE++ ($k=1$)	46.7	832	0.0561
GRPO ($k=4$)	46.8	860	0.0544
RLOO ($k=4$)	44.6	866	0.0515
ReMax ($k=1+1$)	45.1	805	0.0560

Table 1. Comparison on Chat-Arena-Hard (Li et al., 2024). The single-sample REINFORCE++ ($k = 1$) achieves a top-tier score while being more token-efficient than the group-sampling GRPO.

Experimental Results As shown in Table 1, the single-sample REINFORCE++ ($k = 1$) achieves a score of 46.7, statistically tied with the group-sampling GRPO (46.8). However, it produces shorter responses (832 tokens vs. 860), resulting in a higher and more efficient per-token score (0.0561). This result highlights that for general tasks, group sampling ($k > 1$) is not necessary and may even be suboptimal.

Results Analysis Figure 2 shows the training dynamics. GRPO’s reward rises quickly, but its KL divergence also increases rapidly, which is consistent with over-optimizing the reward model. In contrast, REINFORCE++ shows a more stable reward increase with a smaller KL divergence. This behavior is consistent with global normalization improving the stability of advantage estimates and mitigating length-driven reward exploitation (Dubois et al., 2024).

4.2. Reasoning Experiments

Next, we evaluate REINFORCE++ (group-sample, $k > 1$) in its target domain: complex reasoning, where group sampling is standard and reward signals are often sparse (e.g., 0/1). We compare it directly with GRPO ($k > 1$).

4.2.1. LONG REASONING TASK

We use rule-based rewards (RLVR) in mathematical reasoning settings (Guo et al., 2025; Seed et al., 2025).

Analysis on Small-Scale Datasets To test for overfitting, we trained on only 30 questions from AIME-24 and evaluated on AIME-25. As shown in Table 2, GRPO (local norm) reaches a near-perfect 95.0% on the training set but drops to 0.0 Pass@1 on the test set, indicating severe overfitting. In contrast, REINFORCE++ with global norm, while scoring lower on the training set (71.0), shows better generalization (2.5 Pass@1, 40.0 Pass@16).

Pass@N	AIME-24 (Train)		AIME-25 (Test)	
	N = 1	N = 1	N = 1	N = 16
GRPO	95.0	0.0	0.0	0.4
REINFORCE++	71.0	2.5	40.0	

Table 2. Comparison on a small training dataset. GRPO (local norm) overfits completely, while REINFORCE++ (global norm) generalizes.

Figure 3 is consistent with this observation. GRPO (left) quickly fits the training questions in just a few steps, whereas REINFORCE++ (right) improves more gradually, enabled by the more stable global normalization signal.

Logical Reasoning (K&K Puzzles) We also tested on the Knights and Knaves (K&K) puzzles (Xie et al., 2025;

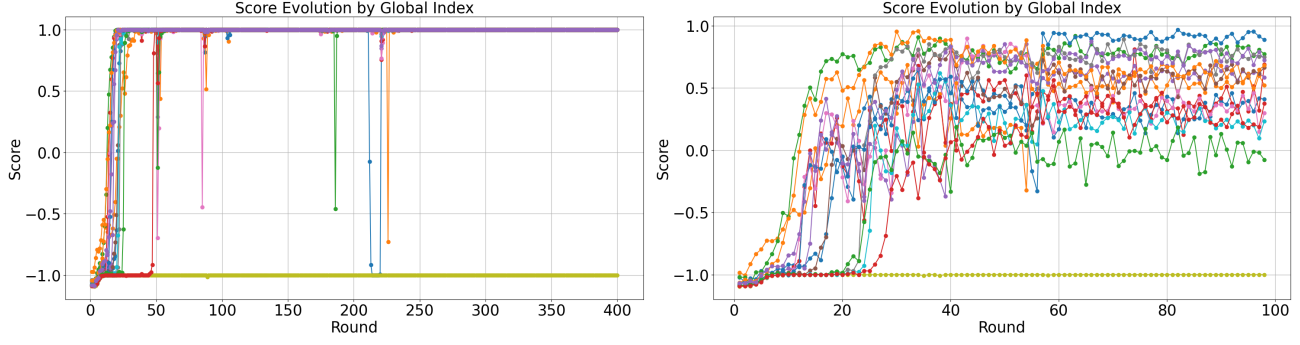


Figure 3. Training curves on a small prompt dataset. Left: GRPO (local norm) immediately overfits. Right: REINFORCE++ (global norm) learns more stably.

2024), where difficulty increases with the number of "people." Figure 4 shows that while GRPO is competitive on easy tasks (2-3 people), its performance collapses on harder, OOD tasks (8 people). REINFORCE++ is more robust, outperforming GRPO on all tasks with four or more people and achieving a much higher average score (62.1 vs. 55.7).

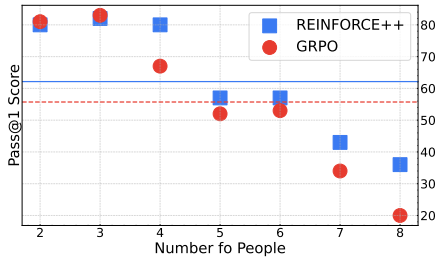


Figure 4. Comparison on logic benchmarks. REINFORCE++'s global normalization is more robust to increasing task difficulty and OOD challenges (8 people).

RL from Zero Setting Finally, we trained a Qwen2.5-Math-Base model from zero on MATH dataset splits (Guo et al., 2025). Table 3 shows that REINFORCE++ again achieves better OOD generalization on the more challenging AIME-24 and AMC-23 datasets, while remaining competitive on the in-distribution MATH-500 test set.

4.3. Multi-step Reinforcement Learning

Finally, to test performance in a complex, multi-step environment, we use the zero-shot agent setup from (Mai et al., 2025), in which a Qwen 2.5 Base 7B model must learn to use Python tools to solve mathematical problems. It is a challenging scenario where group sampling ($k > 1$) and reward reshaping are crucial (see Appendix Section 5).

Experimental Setup We adhere to the training and evaluation protocols of the ZeroTIR environment (Mai et al., 2025). The backbone model is Qwen 2.5 Base 7B, trained using OpenRLHF (Hu et al., 2025a) with datasets from ORZ (Hu

et al., 2025b) and DAPO (Yu et al., 2025). We assess performance on benchmarks including AIME 2024 (Jia, 2024), AIME-2025 (math ai, 2025), HMMT FEB-2024/2025, and CMIMC, using the average@32 metric.

Experimental Result As shown in Table 4, REINFORCE++_{w/ Baseline} achieves the highest average accuracy (24.10) across all benchmarks. It outperforms both GRPO (22.58) and full-critic PPO (21.85) in this tool-use setup. These results suggest that a stable, critic-free approach can be effective for complex agentic tasks. The combination of group-mean reshaping, stable global normalization, and the k_2 KL loss appears to work well in practice.

5. Best Practices

5.1. General Principle

This paper introduces two algorithms: REINFORCE++ and REINFORCE++_{w/ Baseline}, with a key focus on selecting the appropriate variant based on task conditions (notably, whether group sampling with $k > 1$ is feasible and beneficial).

In practice, in open-source RLHF pipelines, REINFORCE++_{w/ Baseline} is often effective in more complex scenarios where group sampling is used. For instance, in multi-turn tool-calling settings, a high proportion of void (i.e., non-informative or incorrect) samples can destabilize training. In such cases, incorporating a baseline via intra-group mean subtraction can improve stability by reducing the impact of uninformative samples. Moreover, this automatic reward reshaping can reduce the need for designing task-specific reward scales. For example, REINFORCE++_{w/ Baseline} supports both 0/1 and $-1/1$ reward schemes, while REINFORCE++ typically works best with symmetric rewards (e.g., $-1/1$) in RLVR tasks.

In contrast, for general-domain tasks where prompt diversity

Pass@N	AIME-24 (OOD) N = 8	AMC-23 (OOD) N = 8	MATH-500 (ID) N = 1
GRPO	18.96	59.22	73.00
REINFORCE++	21.04	60.47	72.00

Table 3. Comparison on RL from Zero. REINFORCE++ improves OOD performance in these settings.

Algorithm	AIME 24	AIME 25	HMMT 2025	HMMT 2024	CMIMC	Avg
GRPO (local norm)	31.66	21.87	16.97	17.70	24.68	22.58
PPO (critic-based)	30.20	21.66	15.00	18.43	23.95	21.85
RF++-Baseline (global norm)	30.83	27.18	17.91	18.95	25.62	24.10

 Table 4. Performance Comparison on complex tool-use benchmarks (average@32). REINFORCE++_{w/ Baseline} outperforms both GRPO (local norm) and PPO (critic-based) models.

and efficiency are paramount, or for tasks where obtaining multiple reward signals for distinct responses is challenging—such as training with Process-Supervised Reward Models (PRMs) or online real-time sampling—we recommend using **REINFORCE++** ($k = 1$). Our experiments (Section 4.1) show improved OOD generalization in these settings.

5.2. Third-Party Validation

The core principle of REINFORCE++, global advantage normalization, has been examined in several independent large-scale reasoning systems, providing additional evidence consistent with its stability and effectiveness.

LitePPO (Liu et al., 2025c), which combines REINFORCE++_{w/ Baseline} with a token-level loss, reports that using a global standard deviation can be more stable than the local standard deviation used in GRPO, and can improve generalization in their evaluated settings.

ScaleRL (Khatri et al., 2025) performed a detailed ablation study on advantage estimation methods in large-scale (16,000 GPU-hour) experiments. They directly compared batch-level normalization in REINFORCE++ with prompt-level normalization in GRPO and found batch-level normalization to be “slightly superior in both compute efficiency and final performance” in their setup.

DLER (Liu et al., 2025b) found that when using truncation to control the output length of an LLM, batch-wise (global) normalization remains stable while group-wise (local) normalization shows declining accuracy. The study provides further evidence that global normalization is more robust to variations in training conditions and reward landscapes.

6. Conclusion

In this paper, we present REINFORCE++, a critic-free RLHF framework designed to enhance training stability by addressing key limitations in existing methods. We identify that prior critic-free algorithms, such as GRPO, rely on *prompt-level (local)* advantage normalization, which is biased for finite group sizes under mild assumptions (Appendix Appendix A) and can be practically unstable.

Our solution, **Global Advantage Normalization**, normalizes advantages across the entire batch, providing a stable estimator whose normalization-induced bias vanishes asymptotically with batch size under standard assumptions. We proposed two variants tailored for different needs:

1. REINFORCE++: For general-domain RLHF, the single-sample ($k = 1$) configuration is computationally efficient and shows improved OOD generalization in our experiments. We also find that its $k > 1$ application is robust for reasoning tasks (Section 4.2).
2. REINFORCE++_{w/ Baseline}: For complex reasoning and agent tasks, this variant combines group-mean reshaping with stable global normalization and a theoretically sound k_2 estimator for KL.

Our empirical results, supported by evidence from multiple third-party systems, are consistent with this specialized approach. REINFORCE++ shows strong generalization in general RLHF. REINFORCE++_{w/ Baseline} improves stability, mitigates overfitting in low-data regimes, and outperforms both GRPO and full-critic PPO in our complex, long-horizon tool-use benchmarks. Overall, our results suggest that a principled critic-free approach can be competitive with traditional PPO while reducing computational overhead.

Impact Statement

Reinforcement learning is increasingly used to steer large language models by optimizing them against a reward signal, including verifiable, programmatic rewards (RLVR) for domains such as math, code, and tool use. The main potential positive impact of this line of work is improved scalability and reproducibility of post-training: verifiable rewards can reduce reliance on human labeling and make optimization targets easier to audit and replicate. By improving the stability and efficiency of critic-free optimization under sparse and brittle rewards, our method can help make RLVR-style training more reliable.

Potential risks include reward gaming when verifiers are incomplete or mis-specified, and dual-use concerns if improved tool-use or problem-solving capabilities are deployed without safeguards. We recommend evaluating beyond verifier scores (e.g., robustness and red-teaming), stress-testing verifiers for exploitability, and using conservative deployment controls (access restriction, monitoring) for systems trained with RL.

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Appendix

A. Proof: The GRPO Advantage Estimator is Biased

A.1. Assumptions and Settings

We observe N rewards r_i for a prompt, and assume the true baseline is θ , such that

$$r_i = \theta + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2), \quad i = 1, \dots, N,$$

with all advantage value ϵ_i independent. Define

$$\bar{\epsilon} = \frac{1}{N} \sum_{j=1}^N \epsilon_j, \quad D = \sqrt{\frac{1}{N} \sum_{j=1}^N (\epsilon_j - \bar{\epsilon})^2}, \quad A_i = \frac{\epsilon_i - \bar{\epsilon}}{D}.$$

We will prove the following:

Theorem A.1. *For any finite $N \geq 2$, the normalized estimator A_i introduces a non-linear dependence on the realized sample, and is biased in the following sense: there does not exist a constant C such that $\mathbb{E}[A_i \mid \epsilon_i] = C \epsilon_i$ holds for all ϵ_i .*

Proof. We prove the claim in the following steps:

Step 1: Numerator Bias We rewrite the numerator:

$$\epsilon_i - \bar{\epsilon} = \left(1 - \frac{1}{N}\right) \epsilon_i - \frac{1}{N} \sum_{j \neq i} \epsilon_j.$$

Since the ϵ_j for $j \neq i$ are zero-mean and independent of ϵ_i ,

$$\mathbb{E}[\epsilon_i - \bar{\epsilon} \mid \epsilon_i] = \left(1 - \frac{1}{N}\right) \epsilon_i.$$

Step 2: Denominator Depends on ϵ_i

(a) Compute $\mathbb{E}[D^2 \mid \epsilon_i]$: By definition,

$$D^2 = \frac{1}{N} \sum_{j=1}^N (\epsilon_j - \bar{\epsilon})^2 = \frac{1}{N} \sum_{j=1}^N \epsilon_j^2 - \bar{\epsilon}^2.$$

Since

$$\bar{\epsilon} = \frac{1}{N} \left(\epsilon_i + \sum_{j \neq i} \epsilon_j \right),$$

and conditioning on ϵ_i keeps the $\epsilon_j, j \neq i$, i.i.d. $\mathcal{N}(0, \sigma^2)$, we obtain:

$$\mathbb{E} \left[\sum_{j=1}^N \epsilon_j^2 \mid \epsilon_i \right] = \epsilon_i^2 + (N-1)\sigma^2,$$

$$\begin{aligned} \mathbb{E}[\bar{\epsilon}^2 \mid \epsilon_i] &= \frac{1}{N^2} \mathbb{E} \left[\left(\epsilon_i + \sum_{j \neq i} \epsilon_j \right)^2 \mid \epsilon_i \right] \\ &= \frac{1}{N^2} \left(\epsilon_i^2 + 2\epsilon_i \cdot \underbrace{\mathbb{E} \left[\sum_{j \neq i} \epsilon_j \right]}_0 + \mathbb{E} \left[\left(\sum_{j \neq i} \epsilon_j \right)^2 \right] \right) \\ &= \frac{\epsilon_i^2 + (N-1)\sigma^2}{N^2}. \end{aligned}$$

Subtracting:

$$\begin{aligned}\mathbb{E}[D^2 \mid \epsilon_i] &= \frac{1}{N}(\epsilon_i^2 + (N-1)\sigma^2) - \frac{\epsilon_i^2 + (N-1)\sigma^2}{N^2} \\ &= \underbrace{\frac{(N-1)^2}{N^2}\sigma^2}_{\alpha} + \underbrace{\frac{N-1}{N^2}\epsilon_i^2}_{\beta} = \alpha + \beta\epsilon_i^2.\end{aligned}\tag{9}$$

(b) $g(\epsilon_i)$ is Not Constant: Let $g(\epsilon_i) = \mathbb{E}[1/D \mid \epsilon_i]$ and $\mu(\epsilon_i) = \mathbb{E}[D^2 \mid \epsilon_i] = \alpha + \beta\epsilon_i^2$.

Using Taylor expansion of $f(x) = 1/\sqrt{x}$ around $x_0 = \mu(\epsilon_i)$:

$$f(x) = \frac{1}{\sqrt{x_0}} - \frac{1}{2} \frac{x - x_0}{x_0^{3/2}} + \frac{3}{8} \frac{(x - x_0)^2}{x_0^{5/2}} + O((x - x_0)^3)$$

Taking the conditional expectation:

$$\begin{aligned}g(\epsilon_i) &= \mathbb{E}[1/D \mid \epsilon_i] = \mathbb{E}\left[\frac{1}{\sqrt{D^2}} \mid \epsilon_i\right] = \mathbb{E}[f(D^2) \mid \epsilon_i], \quad \text{where } f(x) = \frac{1}{\sqrt{x}} \\ &\approx f(\mu(\epsilon_i)) + f'(\mu(\epsilon_i)) \cdot \mathbb{E}[D^2 - \mu(\epsilon_i) \mid \epsilon_i] + \frac{f''(\mu(\epsilon_i))}{2} \cdot \mathbb{E}[(D^2 - \mu(\epsilon_i))^2 \mid \epsilon_i] \\ &= \frac{1}{\sqrt{\mu(\epsilon_i)}} - \frac{1}{2\mu(\epsilon_i)^{3/2}} \cdot \underbrace{\mathbb{E}[D^2 - \mu(\epsilon_i) \mid \epsilon_i]}_{=0} + \frac{3}{8\mu(\epsilon_i)^{5/2}} \cdot \text{Var}(D^2 \mid \epsilon_i) \\ &= \frac{1}{\sqrt{\mu(\epsilon_i)}} + \frac{3}{8} \cdot \frac{\text{Var}(D^2 \mid \epsilon_i)}{\mu(\epsilon_i)^{5/2}}\end{aligned}$$

Since $\mu(\epsilon_i) = \alpha + \beta\epsilon_i^2$ with $\beta > 0$, the first term alone shows that $g(\epsilon_i)$ depends on ϵ_i^2 and hence is not constant.

Step 3: Putting It Together Decomposing A_i ,

$$A_i = \frac{\epsilon_i - \bar{\epsilon}}{D} = \left(1 - \frac{1}{N}\right) \frac{\epsilon_i}{D} - \left(\frac{1}{N} \sum_{j \neq i} \epsilon_j\right) \cdot \frac{1}{D}.$$

For fixed ϵ_i , the conditional distribution of $\sum_{j \neq i} \epsilon_j$ is symmetric about zero, while $1/D$ is always positive. Thus:

$$\mathbb{E}\left[\left(\frac{-1}{N} \sum_{j \neq i} \epsilon_j\right) \cdot \frac{1}{D} \mid \epsilon_i\right] = 0.$$

It follows that

$$\mathbb{E}[A_i \mid \epsilon_i] = \left(1 - \frac{1}{N}\right) \epsilon_i \cdot \mathbb{E}\left[\frac{1}{D} \mid \epsilon_i\right] = \left(1 - \frac{1}{N}\right) \epsilon_i \cdot g(\epsilon_i).$$

Step 4: Concluding the Bias If there existed a constant C such that $\mathbb{E}[A_i \mid \epsilon_i] = C \epsilon_i$ for all ϵ_i , we would have:

$$\left(1 - \frac{1}{N}\right) g(\epsilon_i) \equiv C \quad \Rightarrow \quad g(\epsilon_i) \equiv \frac{CN}{N-1},$$

which contradicts Step 2 since $g(\epsilon_i)$ is not constant. Therefore, for any finite $N \geq 2$, $\mathbb{E}[A_i \mid \epsilon_i]$ cannot be a constant multiple of ϵ_i , and the normalization introduces a non-linear distortion that depends on ϵ_i^2 .

□

B. KL Penalty Design

In non-critic algorithms like REINFORCE++_{w/ Baseline}, a KL divergence term is often added as a separate loss to constrain the policy π_θ from deviating too far from a reference policy π_{ref} . In practice, this expectation is estimated using samples, and three familiar estimators are used.

Given the importance ratio $\delta(y) = \pi_{ref}(y|\cdot)/\pi_\theta(y|\cdot)$ (where π_θ is the sampling policy and π_{ref} is the reference):

- $k_1(y) = -\log \delta(y) = \log \frac{\pi_\theta(y|\cdot)}{\pi_{ref}(y|\cdot)}$
- $k_2(y) = \frac{1}{2}(\log \delta(y))^2 = \frac{1}{2} \left(\log \frac{\pi_{ref}(y|\cdot)}{\pi_\theta(y|\cdot)} \right)^2$
- $k_3(y) = \delta(y) - 1 - \log \delta(y)$

We analyze which of these (k_1 , k_2 , or k_3) is the correct choice when used as a separate loss term.

Theoretical Reverse KL Gradient We are training with samples from π_θ , so we are approximating the **Reverse KL (RKL)**, $D_{KL}(\pi_\theta || \pi_{ref})$. The practical policy gradient for RKL is (more details are in (Liu et al., 2025a)):

$$\nabla_\theta \mathcal{J}_{RKL}(\theta) = \mathbb{E}_{y \sim \pi_\theta(\cdot|x)} \left[\left(\log \frac{\pi_\theta(y|x)}{\pi_{ref}(y|x)} \right) \nabla_\theta \log \pi_\theta(y|x) \right]$$

k1 Analysis We can **exclude** k_1 as a separate KL loss term. Its gradient does not depend on π_{ref} and thus provides no constraining effect. The loss $\mathcal{J}_{k1} = \mathbb{E}_{y \sim \pi_\theta} [\log \pi_\theta - \log \pi_{ref}]$ differentiates to $\nabla_\theta \mathcal{J}_{k1} = \mathbb{E}_{y \sim \pi_\theta} [\nabla_\theta \log \pi_\theta]$, as the π_{ref} term vanishes. (Note: k_1 is used inside the reward for REINFORCE++, but not as a separate loss term).

k2 Analysis The k_2 estimator provides a gradient that is **equivalent to the theoretical Reverse KL gradient**.

The k_2 loss is (noting that $(\log \frac{\pi_\theta}{\pi_{ref}})^2 = (\log \frac{\pi_{ref}}{\pi_\theta})^2$):

$$\mathcal{J}_{k2as\ loss}(\theta) = \mathbb{E}_{y \sim \pi_\theta(\cdot|x)} \left[\frac{1}{2} \left(\log \frac{\pi_\theta(y|x)}{\pi_{ref}(y|x)} \right)^2 \right]$$

Differentiating the term inside the expectation gives:

$$\begin{aligned} \nabla_\theta \left[\frac{1}{2} \left(\log \frac{\pi_\theta}{\pi_{ref}} \right)^2 \right] &= \left(\log \frac{\pi_\theta}{\pi_{ref}} \right) \cdot \nabla_\theta \left(\log \frac{\pi_\theta}{\pi_{ref}} \right) \\ &= \left(\log \frac{\pi_\theta}{\pi_{ref}} \right) \cdot \nabla_\theta (\log \pi_\theta - \log \pi_{ref}) \\ &= \left(\log \frac{\pi_\theta}{\pi_{ref}} \right) \nabla_\theta \log \pi_\theta \end{aligned}$$

When we take the expectation over $y \sim \pi_\theta$, the gradient of the k_2 loss becomes:

$$\mathbb{E}_{y \sim \pi_\theta} \left[\left(\log \frac{\pi_\theta(y|x)}{\pi_{ref}(y|x)} \right) \nabla_\theta \log \pi_\theta(y|x) \right]$$

This matches the practical RKL gradient. Therefore, k_2 is a theoretically well-motivated and stable choice.

k3 Analysis The k_3 gradient is problematic. It is theoretically an estimator for the **Forward KL** ($KL(\pi_{ref} || \pi_\theta)$), not the Reverse KL. There is a mismatch: we are sampling from π_θ , but the Forward KL gradient requires sampling from π_{ref} .

This method, used in GRPO, suffers from two major issues:

- **Extremely High Variance:** When $\pi_\theta(y)$ becomes tiny for a sample y where $\pi_{ref}(y)$ is moderate, the importance weight $\frac{\pi_{ref}(y)}{\pi_\theta(y)}$ in the gradient estimator can explode, leading to unbounded or extremely large variance in practice.

- **Numerical Instability:** It requires computing p/q via $\exp(\log p - \log q)$, which is prone to overflow.

The result explains why methods using k3 require frequent resetting of π_{ref} to prevent the policies from diverging and the k3 estimator from becoming unstable.

Conclusion Based on this analysis, the k_2 estimator is a principled choice for a KL loss term as it correctly and stably estimates the Reverse KL gradient under the sampling distribution. We therefore use the k_2 estimator in our REINFORCE++_{w/ Baseline} algorithm.